

Assignment Sheet 2

Problem 1

Implement the CH Query Routine first based on an already precomputed CH (can be found for Regierungsbezirk Stuttgart on <https://fmi.uni-stuttgart.de/alg/research/stuff/> and for Germany in this Ilias course).

The file format is as follows. A node is described by:

<ID> <OSMID> <Lat> <Lon> <Height> <Level>

The height (elevation) can be ignored, important is the `Level`. A level i node was contracted in the i -th round. An edge is described by:

<SrcID> <TrgID> <Weight> <Type> <MaxSpeed> <EdgeIdA> <EdgeIdB>

Important additional attributes are `EdgeIdA` and `EdgeIdB`, which is case of a short cut point to the two edges that were replaced by it (-1 otherwise).

Test and benchmark your routine against ordinary Dijkstra. Also measure the effect of a simple variant of *Stall-on-Demand*.

Hint: Reordering the nodes according to the level is recommended; also make sure only to look at edges from/to higher levels for good query performance.

Problem 2

Implement your own CH preprocessing routine based on the ideas that were given in the lecture. For the Germany graph and on a 8-core machine, the preprocessing must not take more than 5 minutes (for the Germany graph) and not create more shortcuts than there were original edges in the graph. The file format must be identical to the above.

Compare the query performance of your CH with the ones provided.

For the Germany graph, the CH query times must be less than 10ms.

Problem 3

Submit your program code in ILIAS such that:

- it does not include graph data or compiled files (!)
- to evaluate your code I should only have to:
 - change to your subdirectory
 - copy some `graph.fmi` and `queries.txt` to you directory
 - type `make`

This should create three files:

1. **graph.ch** – your constructed CH
2. **log.txt** – a log of the CH construction process which should include the time to create the CH, time for file I/O, number of created shortcut edges
3. **results.txt** – query results (using your CH) for the queries specified in **queries.txt** (like last time); query time must be in *microseconds*; only integer numbers, no units (e.g.: 42 2025 8494 333 for a query from 42 to 2025 which resulted in a distance of 8494 and within $333\mu s$)